作业

作业题目来自

梁昆淼《力学》 第 4 版 高等教育出版社

例： L01.08 梁昆淼书第 01 章第 08 题

Johns Analytical Mechanics 2nd ed Oxford University Press

例： J02.05 Johns 书第 02 章第 05 题

J03.07† 带 † 表示仅要求荣誉班同学完成

Homework 08, Week 12 2023.11.20 Due 2023.11.27 J19.01, J19.04, J19.05, J19.07, J19.10

★ J19.01

Consider a canonical transformation （正则变换）with constant coefficients, defined by

$$Q_i = \sum_{k=1}^{D} a_{ik}\, q_k \quad \text{or, in matrix form,} \quad [Q] = \mathbb{a}\,[q]$$

$$P_i = \sum_{k=1}^{D} b_{ik}\, p_k \quad \text{or, in matrix form,} \quad [P] = \mathbb{b}\,[p]$$

for $i = 1, 2, \ldots, D$, where the elements a_{ik} and b_{ik} of the matrices $\mathbb{a}$ and $\mathbb{b}$ are all constants.

(a) If the matrix $\mathbb{a}$ is given, find a general expression for the matrix $\mathbb{b}$.

(b) Canonical transformation of this type are of particular importance in quantum theory because the quantum substitution

$p_k \longrightarrow -i\hbar\,\partial/\partial q_k$ implies and is implied by the substitution $P_k \longrightarrow -i\hbar\,\partial/\partial Q_k$. Prove this implication.

(a) If the matrix $\mathbb{a}$ is

$$\mathbb{a} = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}, \quad \text{find the numerical value of each element of the matrix } \mathbb{b}$$

Solution

★ J19.04

A transformation $q, p \longrightarrow Q, P$ is defined by

$$\begin{cases} Q_i = q_i \cos\theta_i + a_i\, p_i \sin\theta_i \\ P_i = -\dfrac{q_i}{a_i} \sin\theta_i + p_i \cos\theta_i \end{cases} \qquad i = 1, 2, \ldots, D$$

where θ_i, with $i = 1, 2, \ldots, D$, are independent, constant parameters, and a_i are given constant having appropriate units.

Use the Poisson-bracket condition to show that this transformation is canonical for any set of θ_i values.

Solution

★ J19.05

Consider two point masses, m_1 and m_2, with position vectors $\vec{r}_i = x_i\,\vec{e}_x + y_i\,\vec{e}_y + z_i\,\vec{e}_z$, $(i = 1, 2)$.

Define the q variables to be $x_1, y_1, z_1, x_2, y_2, z_2$, the p variables to be $p_{1x}, p_{1y}, p_{1z}, p_{2x}, p_{2y}, p_{2z}$

the Q variables to be x, y, z, X, Y, Z and P variables to be $p_x, p_y, p_z, P_x, P_y, P_z$, where

$$\vec{r} = \vec{r}_2 - \vec{r}_1 = x\,\vec{e}_x + y\,\vec{e}_y + z\,\vec{e}_z\,, \quad \vec{R} = \frac{m_2\,\vec{r}_2 + m_1\,\vec{r}_1}{M} = X\,\vec{e}_x + Y\,\vec{e}_y + Z\,\vec{e}_z\,, \quad M = m_1 + m_2, \quad \mu = \frac{m_1\, m_2}{M}$$

$$\vec{p} = \mu \frac{d\vec{r}}{dt}, \quad \vec{P} = M \frac{d\vec{R}}{dt}, \qquad \vec{p}_i = m_i \frac{d\vec{r}_i}{dt} = p_{ix}\,\vec{e}_x + p_{iy}\,\vec{e}_y + p_{iz}\,\vec{e}_z, \quad i = 1, 2$$

(a) Prove that the transformation from $q,\ p$ to $Q,\ P$ is canonical

(b) Show that this is a canonical transformation of type $Q_i = \sum_{k=1}^{D} a_{ik}\, q_k, \quad P_i = \sum_{k=1}^{D} b_{ik}\, q_k,$ as discusse in Problem J19 .01 given above.

and that the standard quantum substitutions

$$\vec{p}_1 \longrightarrow -i\hbar \frac{\partial}{\partial \vec{r}_1}, \quad \vec{p}_2 \longrightarrow -i\hbar \frac{\partial}{\partial \vec{r}_2} \qquad \text{in the } q,\ p \text{ system imply the quntum substitutions}$$

$$\vec{p} \longrightarrow -i\hbar \frac{\partial}{\partial \vec{r}}, \quad \vec{P} \longrightarrow -i\hbar \frac{\partial}{\partial \vec{R}} \qquad \text{in the } Q,\ P \text{ system}$$

Solution

★ J19.07

We know that each product $q_k\, p_k$ for $k = 1, 2, \ldots, D$ has units of action, which are $M\, L^2 / T$.

Demonstrate from the Poisson bracket condition (long form) of canonical transformation that, for all $k = 1, 2, \ldots, D$,

each product $Q_k\, P_k$ after any canonical transformation will have the units of action.

Solution

★ J19.10

A system with one degree of freedom has a Hamiltonian given by

$$H(q, p, t) = \frac{a^2}{2\, m\, p^2} + \frac{m\, \omega^2\, q^2\, p^4}{2\, a^2}$$

(a) Consider the transformation $q,\ p \longrightarrow Q,\ P$ defined as $Q = b\, q\, p^2, \quad P = -\frac{a}{p}$

Assuming the constant a given, for what value of b is this transformation canonical?

(b) Use this transformation with your chosen value of b to write the Hamiltonian in terms of the $Q,\ P$ variables.

(c) Solve the Hamiltonian obtained in part (b) and use the inverse transformation to get the system trajectory, q as a function of t.

What do your learn from this problem?

Solution